Arrest_Data_Tasks

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```r
# Read in cleaned data
prop_2017 <- readRDS("~/Desktop/CSE160_Real_Estate/clean_property_2017.rds")
prop_2017 <- as.data.frame(sf::st_drop_geometry(prop_2017))

str(prop_2017)
```

```
## 'data.frame':    26554 obs. of  9 variables:
##  $ year        : chr  "2017" "2017" "2017" "2017" ...
##  $ borough     : int  2 2 2 2 2 2 2 2 2 2 ...
##  $ tot_sqft    : int  1460 1440 1440 1587 1497 1764 1340 2820 2047 2400 ...
##  $ price       : num  305000 178000 449000 140000 246000 ...
##  $ land_sqft   : int  1330 1306 1306 1622 1694 3525 3525 1293 2356 2000 ...
##  $ yr_built    : int  1899 1899 1899 1899 1899 1899 1899 1952 1901 1993 ...
##  $ bldg_ctgy   : chr  "01 ONE FAMILY DWELLINGS" "01 ONE FAMILY DWELLINGS" "01 ONE FAMILY DWELLINGS"
##  $ bldg_num_ctgy: num  1 1 1 1 1 1 1 1 1 2 ...
##  $ sale_date   : chr  "2017-07-18" "2017-01-19" "2017-07-14" "2017-05-12" ...
```

```r
# Map integer borough codes -> names
prop_2017$borough_code <- as.integer(prop_2017$borough)

prop_2017$borough_name <- prop_2017$borough_code
prop_2017$borough_name[prop_2017$borough_code == 1] <- "Manhattan"
prop_2017$borough_name[prop_2017$borough_code == 2] <- "Bronx"
prop_2017$borough_name[prop_2017$borough_code == 3] <- "Brooklyn"
prop_2017$borough_name[prop_2017$borough_code == 4] <- "Queens"
prop_2017$borough_name[prop_2017$borough_code == 5] <- "Staten Island"

prop_2017$borough <- factor(prop_2017$borough_name)
```

```r
# remove weird / extreme prices (already filtered > 10, but we can also trim top 1%)
q_hi <- quantile(prop_2017$price, 0.99, na.rm = TRUE)
prop_2017 <- prop_2017[prop_2017$price <= q_hi, ]

prop_2017$log_price    <- log(prop_2017$price)
prop_2017$log_tot_sqft <- log(prop_2017$tot_sqft + 1)  # +1 to avoid log(0)
```

```r
arrests_boro_year <- read.csv(
  "NYPD_Arrests_Borough_Year_Summary.csv",
  stringsAsFactors = FALSE
)

crime_boro <- aggregate(
  cbind(total_arrests, felony_arrests, misdemeanor_arrests, violation_arrests) ~ borough,
  data = arrests_boro_year,
  FUN = sum
```

```r
)

crime_boro
```

```
##         borough total_arrests felony_arrests misdemeanor_arrests
## 1         Bronx         47821          19312               28050
## 2      Brooklyn         60046          24684               33547
## 3     Manhattan         50699          19242               30193
## 4        Queens         44433          18187               25073
## 5 Staten Island          9056           3370                5646
##   violation_arrests
## 1               190
## 2              1573
## 3               621
## 4               517
## 5                11
```

```r
# merge on borough name
model_df <- merge(
  prop_2017,
  crime_boro,
  by = "borough"
)

str(model_df)
```

```
## 'data.frame':    26288 obs. of  17 variables:
##  $ borough            : Factor w/ 5 levels "Bronx","Brooklyn",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ year               : chr  "2017" "2017" "2017" "2017" ...
##  $ tot_sqft           : int  1460 1440 1440 1587 1497 1764 1340 2820 2047 2400 ...
##  $ price              : num  305000 178000 449000 140000 246000 ...
##  $ land_sqft          : int  1330 1306 1306 1622 1694 3525 3525 1293 2356 2000 ...
##  $ yr_built           : int  1899 1899 1899 1899 1899 1899 1899 1952 1901 1993 ...
##  $ bldg_ctgy          : chr  "01 ONE FAMILY DWELLINGS" "01 ONE FAMILY DWELLINGS" "01 ONE FAMILY DWELL
##  $ bldg_num_ctgy      : num  1 1 1 1 1 1 1 1 1 2 ...
##  $ sale_date          : chr  "2017-07-18" "2017-01-19" "2017-07-14" "2017-05-12" ...
##  $ borough_code       : int  2 2 2 2 2 2 2 2 2 2 ...
##  $ borough_name       : chr  "Bronx" "Bronx" "Bronx" "Bronx" ...
##  $ log_price          : num  12.6 12.1 13 11.8 12.4 ...
##  $ log_tot_sqft       : num  7.29 7.27 7.27 7.37 7.31 ...
##  $ total_arrests      : int  47821 47821 47821 47821 47821 47821 47821 47821 47821 47821 ...
##  $ felony_arrests     : int  19312 19312 19312 19312 19312 19312 19312 19312 19312 19312 ...
##  $ misdemeanor_arrests: int  28050 28050 28050 28050 28050 28050 28050 28050 28050 28050 ...
##  $ violation_arrests  : int  190 190 190 190 190 190 190 190 190 190 ...
```

```r
n_props_boro <- table(model_df$borough)
model_df$props_in_boro <- as.numeric(n_props_boro[model_df$borough])

model_df$felony_per_prop      <- model_df$felony_arrests      / model_df$props_in_boro
model_df$misdemeanor_per_prop <- model_df$misdemeanor_arrests / model_df$props_in_boro
model_df$violation_per_prop   <- model_df$violation_arrests   / model_df$props_in_boro
model_df$total_per_prop       <- model_df$total_arrests       / model_df$props_in_boro
```

```r
# treat borough as factor
model_df$borough <- factor(model_df$borough)
```

```r
m_base <- lm(
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy + borough,
  data = model_df
)

summary(m_base)
```

```
##
## Call:
## lm(formula = log_price ~ log_tot_sqft + yr_built + land_sqft +
##     bldg_num_ctgy + borough, data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3866  -0.1798   0.1385   0.3952   2.1939
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.263e+01  1.130e-01 111.773  < 2e-16 ***
## log_tot_sqft           3.897e-02  6.626e-03   5.882 4.10e-09 ***
## yr_built              -2.434e-04  5.994e-05  -4.060 4.92e-05 ***
## land_sqft              7.394e-05  3.053e-06  24.223  < 2e-16 ***
## bldg_num_ctgy          1.490e-01  9.223e-03  16.153  < 2e-16 ***
## boroughBrooklyn        5.789e-01  1.917e-02  30.206  < 2e-16 ***
## boroughManhattan       1.338e+00  9.898e-02  13.517  < 2e-16 ***
## boroughQueens          3.430e-01  1.835e-02  18.686  < 2e-16 ***
## boroughStaten Island  1.002e-01  2.086e-02   4.802 1.58e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8886 on 26279 degrees of freedom
## Multiple R-squared:  0.08256,    Adjusted R-squared:  0.08228
## F-statistic: 295.6 on 8 and 26279 DF,  p-value: < 2.2e-16
```

Model 1: total crime intensity

```r
m_crime_total <- lm(
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy + borough +
    total_per_prop,
  data = model_df
)

summary(m_crime_total)
```

```
##
## Call:
## lm(formula = log_price ~ log_tot_sqft + yr_built + land_sqft +
##     bldg_num_ctgy + borough + total_per_prop, data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3866  -0.1798   0.1385   0.3952   2.1939
##
## Coefficients: (1 not defined because of singularities)
```

```
##                        Estimate Std. Error t value Pr(>|t|)    
## (Intercept)           1.263e+01  1.130e-01 111.773  < 2e-16 ***
## log_tot_sqft          3.897e-02  6.626e-03   5.882 4.10e-09 ***
## yr_built             -2.434e-04  5.994e-05  -4.060 4.92e-05 ***
## land_sqft             7.394e-05  3.053e-06  24.223  < 2e-16 ***
## bldg_num_ctgy         1.490e-01  9.223e-03  16.153  < 2e-16 ***
## boroughBrooklyn       5.789e-01  1.917e-02  30.206  < 2e-16 ***
## boroughManhattan      1.338e+00  9.898e-02  13.517  < 2e-16 ***
## boroughQueens         3.430e-01  1.835e-02  18.686  < 2e-16 ***
## boroughStaten Island  1.002e-01  2.086e-02   4.802 1.58e-06 ***
## total_per_prop               NA         NA      NA       NA    
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 0.8886 on 26279 degrees of freedom
## Multiple R-squared:  0.08256,    Adjusted R-squared:  0.08228 
## F-statistic: 295.6 on 8 and 26279 DF,  p-value: < 2.2e-16
```

Model 2: different crime types

```r
m_crime_types <- lm(
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy + borough +
    felony_per_prop + misdemeanor_per_prop + violation_per_prop,
  data = model_df
)

summary(m_crime_types)
```

```
## 
## Call:
## lm(formula = log_price ~ log_tot_sqft + yr_built + land_sqft + 
##     bldg_num_ctgy + borough + felony_per_prop + misdemeanor_per_prop + 
##     violation_per_prop, data = model_df)
## 
## Residuals:
##      Min       1Q   Median       3Q      Max 
## -10.3866  -0.1798   0.1385   0.3952   2.1939 
## 
## Coefficients: (3 not defined because of singularities)
##                        Estimate Std. Error t value Pr(>|t|)    
## (Intercept)           1.263e+01  1.130e-01 111.773  < 2e-16 ***
## log_tot_sqft          3.897e-02  6.626e-03   5.882 4.10e-09 ***
## yr_built             -2.434e-04  5.994e-05  -4.060 4.92e-05 ***
## land_sqft             7.394e-05  3.053e-06  24.223  < 2e-16 ***
## bldg_num_ctgy         1.490e-01  9.223e-03  16.153  < 2e-16 ***
## boroughBrooklyn       5.789e-01  1.917e-02  30.206  < 2e-16 ***
## boroughManhattan      1.338e+00  9.898e-02  13.517  < 2e-16 ***
## boroughQueens         3.430e-01  1.835e-02  18.686  < 2e-16 ***
## boroughStaten Island  1.002e-01  2.086e-02   4.802 1.58e-06 ***
## felony_per_prop              NA         NA      NA       NA    
## misdemeanor_per_prop         NA         NA      NA       NA    
## violation_per_prop           NA         NA      NA       NA    
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
```

```
## Residual standard error: 0.8886 on 26279 degrees of freedom
## Multiple R-squared:  0.08256,    Adjusted R-squared:  0.08228
## F-statistic: 295.6 on 8 and 26279 DF,  p-value: < 2.2e-16
```

K-fold cross-validation

```r
set.seed(160)

kfold_cv <- function(data, formula, K = 5) {
  n <- nrow(data)
  # random fold assignments
  folds <- sample(rep(1:K, length.out = n))

  rmse_vals <- numeric(K)

  for (k in 1:K) {
    train_idx <- which(folds != k)
    test_idx  <- which(folds == k)

    train_dat <- data[train_idx, ]
    test_dat  <- data[test_idx, ]

    fit <- lm(formula, data = train_dat)
    preds <- predict(fit, newdata = test_dat)
    y_true <- test_dat$log_price

    rmse_vals[k] <- sqrt(mean((preds - y_true)^2, na.rm = TRUE))
  }

  mean(rmse_vals)
}
```

```r
aggregate(total_per_prop ~ borough, data = model_df, function(x) length(unique(x)))
```

```
##         borough total_per_prop
## 1         Bronx              1
## 2      Brooklyn              1
## 3     Manhattan              1
## 4        Queens              1
## 5 Staten Island              1
```

```r
aggregate(felony_per_prop ~ borough, data = model_df, function(x) length(unique(x)))
```

```
##         borough felony_per_prop
## 1         Bronx               1
## 2      Brooklyn               1
## 3     Manhattan               1
## 4        Queens               1
## 5 Staten Island               1
```

```r
m_noboro <- lm(
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy,
  data = model_df
)

m_noboro_crime <- lm(
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy +
```

```r
    total_per_prop,
  data = model_df
)

summary(m_noboro)
```

```
##
## Call:
## lm(formula = log_price ~ log_tot_sqft + yr_built + land_sqft +
##     bldg_num_ctgy, data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1167  -0.2108   0.0773   0.4231   2.4501
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.321e+01  1.142e-01 115.703  < 2e-16 ***
## log_tot_sqft   6.948e-02  6.688e-03  10.390  < 2e-16 ***
## yr_built      -4.822e-04  6.087e-05  -7.922 2.43e-15 ***
## land_sqft      5.195e-05  2.994e-06  17.350  < 2e-16 ***
## bldg_num_ctgy  1.781e-01  9.130e-03  19.509  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9108 on 26283 degrees of freedom
## Multiple R-squared:  0.036,  Adjusted R-squared:  0.03585
## F-statistic: 245.4 on 4 and 26283 DF,  p-value: < 2.2e-16
```

```r
summary(m_noboro_crime)
```

```
##
## Call:
## lm(formula = log_price ~ log_tot_sqft + yr_built + land_sqft +
##     bldg_num_ctgy + total_per_prop, data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1188  -0.2086   0.0797   0.4235   2.4230
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     1.319e+01  1.140e-01 115.651  < 2e-16 ***
## log_tot_sqft    6.608e-02  6.687e-03   9.882  < 2e-16 ***
## yr_built       -4.633e-04  6.081e-05  -7.619 2.64e-14 ***
## land_sqft       5.369e-05  2.996e-06  17.922  < 2e-16 ***
## bldg_num_ctgy   1.745e-01  9.125e-03  19.129  < 2e-16 ***
## total_per_prop  1.510e-03  1.651e-04   9.148  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9094 on 26282 degrees of freedom
## Multiple R-squared:  0.03906,    Adjusted R-squared:  0.03888
## F-statistic: 213.6 on 5 and 26282 DF,  p-value: < 2.2e-16
```

```r
set.seed(160)
rmse_noboro <- kfold_cv(
  model_df,
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy,
  K = 5
)

set.seed(160)
rmse_noboro_crime <- kfold_cv(
  model_df,
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy + total_per_prop,
  K = 5
)

rmse_noboro
```

```
## [1] 0.9101806
```

```r
rmse_noboro_crime
```

```
## [1] 0.908784
```

Out of sample performance

```r
# baseline
rmse_base <- kfold_cv(
  model_df,
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy + borough,
  K = 5
)

# with total crime
rmse_crime_total <- kfold_cv(
  model_df,
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy + borough +
    total_per_prop,
  K = 5
)

# with crime types
rmse_crime_types <- kfold_cv(
  model_df,
  log_price ~ log_tot_sqft + yr_built + land_sqft + bldg_num_ctgy + borough +
    felony_per_prop + misdemeanor_per_prop + violation_per_prop,
  K = 5
)

rmse_base
```

```
## [1] 0.8874289
```

```r
rmse_crime_total
```

```
## [1] 0.8877406
```

```r
rmse_crime_types
```

```
## [1] 0.8865737
```

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```r
boro_summary <- aggregate(
  cbind(price, log_price) ~ borough,
  data = model_df,
  FUN = mean
)

# Calculate average arrests-per-property for each borough
boro_summary$total_per_prop <- tapply(
  model_df$total_per_prop,
  model_df$borough,
  mean
)

# Build plotting data frame
df_plot <- data.frame(
  borough   = boro_summary$borough,
  log_price = boro_summary$log_price,
  log_crime = log10(boro_summary$total_per_prop + 1)
)

df_plot
```

```
##         borough log_price log_crime
## 1         Bronx  12.92664 1.2095304
## 2      Brooklyn  13.48021 0.9824436
## 3     Manhattan  14.23821 2.7866317
## 4        Queens  13.23810 0.7130108
## 5 Staten Island  13.01289 0.4271512
```

```r
label_df <- df_plot

# move labels slightly above points
label_df$log_crime_label  <- label_df$log_crime
label_df$log_price_label  <- label_df$log_price + 0.04

ggplot(df_plot, aes(x = log_crime, y = log_price)) +
  geom_point(size = 3, color = "steelblue") +

  # trend line
  geom_smooth(method = "lm", se = FALSE,
              color = "gray40", linetype = "dashed") +

  geom_text(
    data = label_df,
    aes(x = log_crime_label, y = log_price_label, label = borough),
    size = 5,
    vjust = 0
  ) +
  coord_cartesian(
    ylim = c(min(df_plot$log_price) - 0.05,
             max(df_plot$log_price) + 0.15),
```

```
    clip = "off"
  ) +
  theme_minimal(base_size = 15) +
  theme(
    panel.grid.minor = element_blank(),
    plot.margin = ggplot2::margin(10, 40, 10, 10)  # <-- key fix
  ) +
  labs(
    title = "Borough-level Crime vs Mean Property Price",
    x = "log10(Average Total Arrests per Property + 1)",
    y = "Mean Log Sale Price"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```